Caterina Morelli

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Profile

Research Fellow at the University of Valle d'Aosta, I recently earned my Ph.D. in Economics, Statistics and Data Science at University of Milan-Bicocca. My research combines spatial and spatiotemporal methods to study territorial heterogeneity and dynamic in energy use, GHG emissions, corporate sustainability, and tourism flows. I leverage advanced statistical and econometric tools, including spatial and spatiotemporal panel models, difference-in-differences frameworks, event-study analysis, GLMs/GAMs, spline-based spatiotemporal interactions, hierarchical clustering, and gravity models.

Education

Ph.D. **Economics, Statistics and Data Science** — Nov 2021 – Feb 2026
University of Milano-Bicocca, Milan, Italy
Supervisors: Prof. Paolo Maranzano, Prof. Philipp Otto (University of Glasgow)
Thesis: "Sustainability by Proximity: How Geographic Closeness Drives Corporate and Regional Carbon Emissions" (available online)
PhD with honours

M.Sc. **Statistical, Financial and Actuarial Sciences** — Nov 2018 – Oct 2020
Alma Mater Studiorum – University of Bologna, Rimini, Italy
Supervisor: Prof. Enrico Supino
Thesis: "The impact of sustainability awards on firm's market value: the case of SEAL Business Award"
110/110 with honours

B.A. **Economics and Commerce** — Sep 2015 – July 2018
Marche Polytechnic University, Ancona, Italy
Supervisor: Prof. Francesco Maria Chelli
Thesis: "Wellbeing is not evaluated in euros: correlation between Pil and Bes"
110/110 with honours

Visiting Periods

University of Valencia, ES	Apr - May 2026
University of Glasgow, UK	Apr - Sep 2024
Bocconi University, Milan, Italy	Oct 2023 – Jan 2024
Erasmus+: Universitat Jaume I, Castellon de la Plana, Spain	Sept 2017 - Jan 2018

Other educational experience

Postgraduate Course: Network Econometrics, SIdE, Ca' Foscari University of Venice	July 2025
Summer School: Generalized Additive Models in R, Ca' Foscari University of Venice (with Simon N. Wood, University of Edinburgh)	June 2025
Interdiscipinary course: Introduction to grant writing, UniMiB	Apr 2025
PhD Course: Kalman Filter and State Space Models, University of Padua (with Siem Jan Koopman, VU University Amsterdam)	June 2024
Interdiscipinary course: Literature review: managing and evaluating your bibliography, UniMiB	Apr 2024
Summer School: Macroeconomic Forecasting and Analysis in the Machine Learning Era, SIdE (Perugia, Italy)	July 2023

PhD Course: Multivariate analysis, Bocconi University — Oct 2023

PhD Course: Machine Learning, Bocconi University — Oct 2023

PhD Course: Time Series Analysis using UCM and State Space Models, University of Aveiro, Portugal (online) — May 2023

Work Experience

Research fellow - University of Valle d'Aosta — Dec 2025 - ongoing
Project SMARTDECO: Energy and Tourism in Fragile Mountain Areas
Responsible Prof. Consuelo Rubina Nava

Research Assistant

University of Milano-Bicocca
WebSite Premia: Empirical analysis of Italian firms' performance — Dec 2024 - June 2025
Responsibles: Prof Piergiovanna Natale, Prof Valeria Gattai

CIPN Care Project: Analysis of hospital data in Stata — Feb – July 2023
Responsible: Prof Paola Alberti

Teaching Assistant

Bocconi University
Statistics – Bachelor's degree – Prof. Raffaella Piccarreta — 2024/25

University of Milano-Bicocca
Econometrics (in English) – Master's degree – Prof. Luca Stanca — 2024/25
Statistics II – Bachelor's degree – Prof. Riccardo Borgoni — 2024/25
Statistics II – Bachelor's degree – Prof. Riccardo Borgoni — 2023/24
Statistics II – Bachelor's degree – Tutoring for student with SLD — 2022/23

University of Milan (Statale)
Statistics – Bachelor's degree – Prof. Daniele Spinelli — 2024/25
Statistics – Bachelor's degree – Prof. Andrea Ariu — 2023/24
Statistics (in English) – Bachelor's degree – Prof. Silvia Salini — 2022/23
Mathematics – Bachelor's degree – Prof. Giovanni Puccetti — 2022/23
Statistics – Bachelor's degree – Prof. Chiara Tommasi — 2022/23

Subject Expert (Cultore della materia)

University of Valle d'Aosta — 2026/27
Quantitative Methods for Economics and Management; Statistics and Data Analysis; Economic Statistics; Green Economy and Analytics.

University of Milano-Bicocca — 2023 – 2025
Statistics II.

Other Professional Experience

Analyst, Deloitte Consulting S.r.l. — Dec 2020 – Oct 2021
Monitoring and improvement of IT systems for investment and insurance products

Volunteer, Giramundo Project (AIESEC) — July – Aug 2019
Centro Educacional Dom Bosco, Natal, Brazil

Conferences and Workshops

ASA, University of Teramo, Italy* — Sept 2026
ERSA, Sofia University, Bulgaria* — Aug 2026

EcoMod, University of Luxembourg, Luxembourg* Aug 2026
SIS-FEN 2026, Sapienza University of Rome, Italy* June 2026
Milan PhD Economic Workshop, Bocconi University, Italy* May 2026
Seminar, University of Valencia, Spain* May 2026
ICES SIS, University of Bari, Italy* Feb 2026
IWEEE SIdE, Marche Polytechnic University, Italy◇ Jan 2026
AISRe, University of Chieti-Pescara, Italy* Sept 2025
IES, University of Padua, Bressanone-Brixen*† June 2025
ICEEP (CefES), University of Milano-Bicocca, Italy* June 2025
SEEDS, Catholic University of the Sacred Heart, Milan, Italy* May 2025
ReLunch Seminar, University of Milano-Bicocca, Italy* May 2025
SEA, Mohammed V University in Rabat, Morocco§* May 2025
IAERE, Roma Tre University, Rome, Italy* Feb 2025
CfeCms, King's College London, UK*‡ Dec 2024
Milan PhD Economic Workshop, University of Milano-Bicocca, Italy* Sept 2024
MBC2, University of Catania, Italy◇ Aug 2024
METMA XI, Lancaster University, UK◇ July 2024
ICEEP (CefES), ETH Zürich, Switzerland* June 2024
Statistics Seminar, University of Glasgow, UK* June 2024
PhD Seminar, University of Milano-Bicocca, Italy* Feb 2024
IWEEE SIdE, Free University of Bozen-Bolzano, Italy* Jan 2024
ERSA Congress, University of Alicante, Spain* Aug 2023
Milan PhD Economic Workshop, University of Milan (Statale), Italy* June 2023

*Presenter, §Discussant, †Invited session, ‡Session organizer, ◇Poster.

Research Projects

SMARTDECO – Energia e Turismo in Aree Montane Fragili: Strategie e metodologie per il Decoupling 2025 – 2027
Research project funded under the "Aggregazioni R&S – Transizione ecologica" call (FESR Valle d'Aosta 2021/2027). Università della Valle d'Aosta. Scientific coordinator: Prof.ssa Consuelo Rubina Nava.

Website Premia 2024
Research collaboration on the evaluation of Italian firms' website quality and its relationship with firms' performance. Department of Economics, Quantitative Methods and Business Strategies, University of Milano-Bicocca. Scientific coordinators: Prof. Piergiovanna Natale, Prof. Valeria Gattai.

CIPN CARE – Peripheral neuropathy from chemotherapy: why should we care? 2022 – 2023
Collaboration within the research project of the Department of Medicine and Surgery, University of Milano-Bicocca. Scientific coordinator: Dott.ssa Paola Alberti.

Working Papers

- **C. Morelli**, C. R. Nava. *Tourism Composition Matters: Spending Capacity and GHG Emissions in European NUTS2 Regions. (Under Review)*
- **C. Morelli**, P. Maranzano, R. Mattera P. Otto. *Cluster-wise spatiotemporal panel models for global and local spillovers identification.* Preliminary draft available upon request
- M. Longhi, **C. Morelli** (2025). *The role of Chinese Panda Ambassadors in International Trade.* DEMS Working Paper: 550 *(Under Review)*

Peer-reviewed Articles

1. P. Maranzano, R.Mattera, **C. Morelli**, G. Scepi (2026). Forecast reconciliation of agricultural GHG emissions in Europe with fuzzy clustering. *Annals of Operations Research* doi.org/10.1007/s10479-026-07363-y
2. **C. Morelli**, P. Maranzano, P. Otto (2026). Spatiotemporal clustering of GHGs emissions in Europe: exploring the role of spatial component. *Spatial Statistics.* doi.org/10.1016/j.spasta.2026.100960

3. S. Boccaletti, P. Maranzano, **C. Morelli**, E. Ossola (2025). ESG Performance and Stock Market Responses to Geopolitical Turmoil: evidence from the Russia-Ukraine War. *Economic Modelling* doi.org/10.1016/j.econmod.2025.107380

4. **C. Morelli**, S. Boccaletti, P. Maranzano, P. Otto (2025). Multidimensional spatiotemporal clustering – An application to environmental sustainability scores in Europe. *Environmetrics*. doi.org/10.1002/env.2893

5. R. Borgoni, M. Denova, P. Maranzano, **C. Morelli** (2023). Assessing geographical differences of the COVID-19 impact on fuel consumption: the case of Italy. *Letters in Spatial and Resource Sciences*. 10.1007/s12076-023-00361-9

Short paper and Book of abastracs

1. **C. Morelli** (2026). Tourist Spending Capacity and Greenhouse Gas Emissions: Evidence from European NUTS2 Regions. *Book of short papers.* ICES 2026 - 4th International Conference on Economic Statistics

Reviewer for peer-reviwed journals

Economic Modelling
Computational Economics
Annals of Operations Research
International Journal of Climate Change Strategies and Management
Humanities and Social Sciences Communications
Economics of Disaster and Climate Change
Scientific Reports
Communications in Statistics - Theory and Methods.

Scholarships

Erasmus+ scholarship for research period abroad (2024)
Ph.D. 4-year scholarship, University of Milano-Bicocca (2021–2025)
Erasmus+ scholarship for study period abroad (2017)

Languages

Italian (native)
English (fluent)
Spanish (fluent)
French (intermediate)

Software & Programming

Advanced: Microsoft Office / 365, R, LaTeX
Intermediate: VBA, Stata, Python, html
Basic: MySQL, Gretl, Matlab, Dynare

`SCDA` **R package contribution:** HSTC functions for hierarchical spatiotemporal cluster analysis of high-dimensional data; available in the upcoming release SCDA package documentation (CRAN).